Hall Ticket Number:

MR-21

Code: B415CS3S

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. IV Semester Supplementary Examinations, July 2025

Database Management Systems

(Common to CSE and IT)

Time: 3 Hours | ***Answer ALL Questions*** | Max. Marks: 70

Part-A (10 x 2 = 20 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	State the purpose of database systems.	2	1	1	1
1. b)	Name the applications of database system.	2	1	1	1
1. c)	List out the standard operations on sets.	2	1	2	1
1. d)	Define Join and name various types of join operations.	2	1	2	1,2,3
1. e)	What are functional dependencies?	2	1	3	1,2,3
1. f)	Mention the problems caused by redundancy.	2	2	3	1,2,3
1. g)	Why does a DBMS interleave transactions?	2	2	4	1,2
1. h)	Show the technique used in log-based recovery.	2	2	4	1,2,3
1. i)	Summarize the characteristics of a B+ tree.	2	2	5	1,2
1. j)	Label the main alternatives for what to store as a data entry in an index.	2	2	5	1,2

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Draw the architecture of a DBMS and describe the components of database architecture.	5	2	1	1
2. b)	Examine the integrity constraints, that a DBA or user can specify in the relational model.	5	4	1	1,2,3
	OR				
2. c)	Answer each of the following questions. The questions are based on the following relational schema: Emp (eid: integer, ename: string, age: integer, salary: real) Works (eid: integer, did: integer, peLtime: integer) Dept (did: integer, dname: string, budget: real, managerid: integer) 1. Give an example of a foreign key constraint that involves the Dept relation. What are the options for enforcing this constraint when a user attempts to delete a Dept tuple? 2. Write the SQL statements required to create the preceding relations, including appropriate versions of all primary and foreign key integrity constraints. 3. Define the Dept relation in SQL so that every department is guaranteed to have a manager. 4. Write an SQL statement to add John Doe as an employee with eid = 101, age = 32 and salary = 15,000. 5. Write an SQL statement to give every employee a 10 percent raise.	5	3	1	1,2,3
2. d)	Inspect the mechanism for creating and modifying the relations using SQL.	5	4	1	1,2,3
3. a)	Explain the task of the following operators: a) selection b) projection c) union d) cross-product and e) difference	5	2	2	1
3. b)	Interpret the approach of aggregate operators supported by SQL.	5	3	2	1,2,3
	OR				

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3. c)	Illustrate the features of tuple relational calculus and Domain relational calculus.	5	2	2	1
3. d)	Analyse the utilization of union, intersect, and except operations.	5	4	2	1,2
4. a)	Define Armstrong's Axioms. Elucidate the closure of a set of dependencies.	5	2	3	1
4. b)	Assess the functionality of first, second and third normal forms.	5	3	3	1,2
	OR				
4. c)	Discuss the concept of Trivial and Non-Trivial functional dependencies	5	2	3	1,2
4. d)	Infer the strategy followed in BCNF and fourth normal form.	5	4	3	1,2
5. a)	Diagnose the important properties of transactions to maintain data in the face of concurrent access and system failures.	5	5	4	1,2
5. b)	Demonstrate the role of strict two-phase locking protocol.	5	2	4	1,2
	OR				
5. c)	Elaborate the impact of various anomalies due to interleaved execution.	5	2	4	1,2
5. d)	Outline the implementation of timestamp-based protocols and validation- based protocols.	5	2	4	1,2
6. a)	Show the process of Hash-Based indexing with a suitable example.	5	2	5	1,2
6. b)	Label the ISAM index structure and interpret the operations performed on ISAM.	5	3	5	1,2
	OR				
6. c)	With an appropriate example, discuss the procedure used in Tree-Based indexing.	5	2	5	1,2
6. d)	How does a B+ tree index handle search, insert, and delete? Explain.	5	5	5	1,2

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome

Hall Ticket Number:

MR-21

Code: B415CB3S

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. IV Semester Supplementary Examinations, July 2025

Introduction to Database Management Systems

(Common to CSB, CSM and CSD)

Time: 3 Hours | ***Answer ALL Questions*** | Max. Marks: 70

Part-A (10 x 2 = 20 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	Define schema and instance?	2	1	1	1
1. b)	List Various Types of Attributes?	2	2	1	1
1. c)	Discuss the use of Rename operation?	2	2	2	1
1. d)	Write about project operation relational algebra?	2	2	2	1
1. e)	Write about functional dependency?	2	2	3	1,2
1. f)	Define redundancy and list the problems with it?	2	1	3	1,2
1. g)	Demonstrate conflict serializability?	2	3	4	1,2
1. h)	Explain about transaction stages?	2	2	4	1,2
1. i)	Discuss about data on external storage?	2	2	5	1,2
1. j)	Define tree indexing?	2	1	5	1,2

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Explain the Architecture of the database?	5	1	1	1
2. b)	Compare different data modes?	5	2	1	1,2
	OR				
2. c)	Discuss about different types of integrity constraints?	5	3	1	1,2
2. d)	Explain about the E-R diagram concept?	5	3	1	1,2
3. a)	Illustrate different set operations in relational algebra with an example?	5	3	2	1
3. b)	Discuss about domain relational calculus in detail?	5	3	2	1
	OR				
3. c)	Explain all the aggregate functions with example?	5	6	2	1,2
3. d)	Define trigger and explain all the three parts of it with an example?	5	2	2	1,2
4. a)	Illustrate dependency? And explain the problems caused by it?	5	4	3	1,2,3
4. b)	Define normalization and explain 1st and 2nd normal forms with an example?	5	5	3	1,2,3
	OR				
4. c)	Define and Compare BCNF with 3rd Normal forms?	5	4	3	1,2,3
4. d)	Illustrate multivalued dependency and 4th normal form with an example?	5	3	3	1,2,3
5. a)	List and explain ACID properties in detail?	5	2	4	1,2
5. b)	Explain about timestamp-based protocol?	5	2	4	1,2
	OR				
5. c)	Discuss about Two-Phase locking protocol in detail?	5	2	4	1,2
5. d)	What is recovery and explain about log-based recovery with example?	5	2	4	1,2

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6. a)	Explain about B+ tree indexing with an example?	5	2	5	1.2
6. b)	Explain about cluster, primary and secondary indexes with example?	5	2	5	1,2
	OR				
6. c)	Explain about hash based indexing with an example?	5	2	5	1,2
6. d)	Explain in detail and ISAM?	5	2	5	1,2

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome